

Ariel Pratama Lesmana

GenAI Engineer • Bali, Indonesia (remote)

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Summary

GenAI engineer building computer vision and AI for robots in production. 3+ years building production systems.

Experience

GenAI Engineer

Senserbot PTE. LTD

Jun 2026 — Present

Remote

- **Built computer-vision OCR for field photos** — reads niche numbers and Chinese and English names from photos taken on site, with a dashboard to correct misreads.
- **Built the patrol logic for museum security robots** — a state machine, POIs and routes editable over an API, floor-to-floor lift rides through the elevator vendor's API with an audit trail, and vision checks on an edge PC.
- **Shipped on-robot computer vision** — artifact-intrusion and unattended-object detection (YOLO, hand tracking, overlap checks) that raises alerts with evidence images.
- **Built operator dashboards and CMSs** — robot controls, POI and route editors, scan-to-scan anomaly comparison for a columbarium, and a tour-guide content CMS now in production.
- **Taking over a speech-to-answer GenAI service** for tour-guide robots: scoped the handover and started RAG work on its answers.
- **Built the team's kanban board with an MCP server**, so engineers and their AI agents work from the same cards.

Stack: Python, FastAPI, OpenCV, YOLO, MediaPipe, ROS, Intel RealSense, React, Next.js, Docker, AWS S3, MCP

Fullstack Developer

Insignia

Jun 2025 — Jun 2026

Remote

- **Built a full-stack dashboard system** improving data visibility and operational efficiency for clients.
- **Designed RESTful APIs** for reliable backend integrations across multiple client systems.
- **Integrated GenAI models** into production — pushed latest improvements live without regressions.
- **Optimized DB queries** for measurable response-time gains and better app performance.

Stack: Next.js, Node.js, REST API, Azure, PostgreSQL, GenAI, LLM

Software Engineer

Etherval IT Consultancy

Aug 2023 — May 2025

Surabaya, Indonesia

- **Led ERP development** used daily by 10–20 users; cut lag in key flows.
- **Integrated AWS Lambda, ECR, ECS, S3** — slashed debugging time, hit 80% reliability for AI functions.
- **Engineered an AI fitness app** with CNNs and Google ML libs to track exercise form — 95%+ accuracy.
- **Optimized report-gen query perf** across ~10k records on MariaDB, PostgreSQL, MongoDB.
- **Introduced automated testing** — reduced deployment errors to 2%; integration tests on every branch.
- **Built CI/CD via GitHub Actions** — production deployment success rate climbed to 90%.

Stack: Spring Boot, NestJS, FastAPI, AWS, Docker, TensorFlow, PostgreSQL, MongoDB, MariaDB, GitHub Actions

Skills

AI/ML: GenAI, LLM Integration, RAG, sentence-transformers, Prompt Engineering, NLP, TensorFlow, CNN, Computer Vision, YOLO, OpenCV, MediaPipe, MCP

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML/CSS

Backend: Node.js, NestJS, FastAPI, Spring Boot, REST API

Frontend: Next.js, React

Databases: PostgreSQL, MongoDB, MariaDB, Supabase, ChromaDB

Infra: Docker, AWS, Azure, GitHub Actions, CI/CD, Git, Linux, ROS

Education & Certifications

Institut Sains dan Teknologi Terpadu Surabaya

Bachelor of Informatics, Computer Science

Jul 2019 — Jul 2023

Surabaya, Indonesia

Coursework: Data Structures, Big Data, Machine Learning, Deep Learning, Artificial Intelligence, Generative Algorithm, Data Analytics, Data Mining

Intro to Machine Learning, Kaggle (Jan 2025) • AWS Academy Graduate — Data Analytics, Amazon Web Services (2023)

Indonesian (Native), English (Professional)